

Complete Infrastructure, AWS & EMQX Cloud Master Report

Project: 4Layers / SmartNest Smart Home IoT Platform | Date: July 2026

1. System Architecture Overview

The system operates on a cloud-native microservice architecture designed for real-time IoT hardware control with zero maintenance overhead:

Component	Technology / Service	Function
Mobile Client	React Native (Expo)	Cross-platform user app for device & room control
Backend API	AWS App Runner (FastAPI)	Auto-scaling production API container
Database	Amazon RDS PostgreSQL	Managed relational DB with daily auto-backups
MQTT Broker	EMQX Cloud Serverless	TLS Encrypted real-time hardware communication
IoT Hardware	ESP32 Nodes / Relays	Smart appliance control switches

2. AWS Infrastructure & Active Credits Analysis

The AWS account has **\$120.00 Active Promotional Credits** applied. Below is the usage breakdown:

Metric	Status / Value	Details
Total Credits Granted	\$120.00 USD (~ █10,000 INR)	Applied to AWS Billing Account
Credit Expiry Date	July 28, 2027	Valid for 1 full year
Amount Used Till Date	\$0.64 USD (~ █53 INR)	One-time resource setup & build cost
Remaining Free Credits	\$119.36 USD (~ █9,950 INR)	Available balance for upcoming billing
Estimated Monthly Usage	~\$5.00 / month (~ █400 / mo)	App Runner compute + RDS Free Tier
Client Net Cost (Year 1)	█0.00 / Month (100% FREE)	100% covered by active AWS credits

3. EMQX Cloud MQTT Broker Metrics

The IoT hardware broker is running on EMQX Serverless Cloud with an active zero-spend safety cap:

Metric Name	Account Limit	Current Usage	Safety Status
Spend Limit	\$0.00 / Month	\$0.00	Zero billing risk guaranteed
Session Minutes	1,000,000 / month (Free)	2,350 minutes	99.77% Free Quota Available
Data Traffic	1.0 GB / month (Free)	0.00 GB	100% Free Quota Available
Concurrent Sessions	1,000 Devices	Ready for scale	Supports up to 1,000 IoT nodes

4. Executive Cost Summary & Year 2 Options (July 2027 Onwards)

Time Period	Infrastructure Setup	Monthly Expense	Total Yearly Expense
Year 1 (Now – July 2027)	AWS App Runner + RDS + \$120 Credits	■0.00 / Month	■0.00 / Year
Year 2 (Option A)	Continue Full AWS Enterprise	~ ■2,000 / Month	~ ■24,000 / Year
Year 2 (Option B - Rec.)	Single High-Speed Cloud VPS Server	~ ■600 / Month	~ ■7,200 / Year

Note for Client: Year 1 is 100% covered by AWS credits with zero out-of-pocket costs. In Year 2, Option B reduces ongoing operational expenses by 70% while supporting thousands of active IoT devices.